

Work Experience

OpenCity Labs - Startup

Remote

AI & Machine Learning ENGINEER - R&D

August 2025 - Current

- Fine-tuned a Transformer-based NER model for multilingual PII anonymization, paired with a reversible in-memory mapping for lossless re-identification; served it as a stateless, asynchronous FastAPI microservice on AWS EC2 handling 50k+ daily inputs.
- Diagnosed and fixed a cross-lingual generation failure where 35-40% of non-Italian queries drifted to Italian, using a spaCy language-detection guard with an LLM fallback; shipped to 10+ clients.
- Built a hybrid retrieval system fusing dense embeddings with full-text search via reciprocal rank fusion, resolving legal and regulatory queries where vector search missed acronyms and keyword search missed semantics.
- Containerized services and automated build, test, and deployment with Docker and GitLab CI/CD, cutting monthly infrastructure cost by 85% and per-instance marginal cost by 90%.

Python, PyTorch, Transformers, Fine-tuning, NER, Model Deployment, FastAPI, Docker, AWS EC2

Eurecat Technology Center

Barcelona, Spain

INTERN / AI ENGINEER

April 2024 - June 2024

- Implemented a token-level uncertainty estimation framework for planning tasks by analyzing log-probabilities, benchmarking performance across proprietary (GPT-4o) and locally quantized models (Llama 3.1 via llama.cpp).
- Developed a RAG system utilizing *gte-large* embeddings to retrieve context-aware user preferences, significantly reducing model hallucination in recurrent decision-making scenarios.
- Prototyped a multimodal agent integrating LLaVA for visual reasoning tasks and validated the uncertainty reconstruction approach across multiple simulated environments.

Python, PyTorch, Hugging Face, OpenAI API, llama.cpp, Ollama

Projects

COVID-19 Lung Ultrasound Images classification - Medical Imaging Diagnostic

[slides | report | code]

- Designed a multi-stage pipeline for COVID-19 LUS severity scoring (0-3): a fine-tuned ResNet18 frame classifier, a binary SVC uncertainty detector on softmax outputs, and a t-SNE-based similarity module retrieving nearest neighbours for low-confidence predictions.
- Trained and tested on 47k anonymized frames from 14 patients, with patient-level splits to prevent data leakage.

Python, PyTorch, scikit-learn, Pandas, NumPy, Azure, Computer Vision

Joint Intent Detection and Slot Filling - Natural Language Understanding

[report | code]

- Compared 4 architectures for joint intent detection and slot filling, from fine-tuned pre-trained transformers (BERT, ERNIE) to recurrent models built from scratch (BiLSTM, Encoder-Decoder).
- Consistently improved baseline results by at least +2% on ATIS and +13% on SNIPS datasets.

Python, PyTorch, NLP (BERT/ERNIE/LSTM)

Domain Adaptation / Transfer Learning - Deep Learning - Team Project

[code]

- Built an unsupervised domain adaptation model based on ResNet34 with a custom adaptation layer using direct 3rd-order and grouped 4th-order statistics to bridge the synthetic-to-real gap.
- Achieved +11.53% improvement over non-adapted baseline on the Adaptope dataset.

Python, PyTorch, Google Colab

Education

University of Trento

Trento, Italy

Master's Degree in Artificial Intelligence Systems

September 2021 - Graduated March 2025

- **Thesis:** Exploring the Use of LLMs for Agent Planning: Strengths and Weaknesses. [report | code]
- **Relevant Courses:** Machine Learning, Deep Learning, Natural Language Understanding, Automated Planning, Law & Ethics in AI, Fundamentals of AI (Reasoning and Planning), Signal Processing, AI for Finance, High Performance Computing.

University of Trento

Trento, Italy

Bachelor's Degree in Computer Science

September 2017 - Graduated June 2021

- **Core Topics:** Algorithms & Data Structures, Software Engineering, Database Systems, Operating Systems, Computer Architectures, Distributed Networks.

Skills

Languages	Python, JavaScript, C/C++, SQL, Bash, TeX
ML & Data	PyTorch, Hugging Face, Transformers, Model Fine-tuning, Training & Benchmarking, Computer Vision, NLP, LLM, scikit-learn
DevOps & Cloud	Git, Docker, GitLab CI, FastAPI, Model Deployment, AWS EC2, Grafana, Prometheus
Spoken Languages	English - Professional, Italian - Native